# Coding agent benchmark

7 harness/model pairs. Twelve curated tasks. 84 scored attempts.

## Objective

Measure task completion, execution time and cost across the recorded coding agent configurations. Compare tools for the selected workload.

## Task selection

The CPU-only subset contains six easier tasks with expert estimates of 0.75–2 hours and six harder tasks estimated at 4–8 hours. Every pair receives the same tasks.

Twelve tasks cover security, routing geometry, music constraints, build artifacts, cryptanalysis, payments, storage, live systems, graph performance and UI behavior.

## Study scope

The unit of comparison is the configured harness/model pair. The dataset is curated, with one attempt per pair and task. Scores describe completion on these twelve tasks.

# Methodology

## Execution and verification

Harbor starts each task environment and installs the pinned harness. The runner supplies the task and model, requests medium reasoning effort, and executes one trial at a time. The agent timeout multiplier is 0.25; an eight-hour task receives a two-hour agent limit.

A separate verifier checks artifacts and behavior after execution. Reward 1 means resolved. Each pair has twelve scored attempts. Runtime failures and provider refusals count as unresolved.

## Configuration and accounting

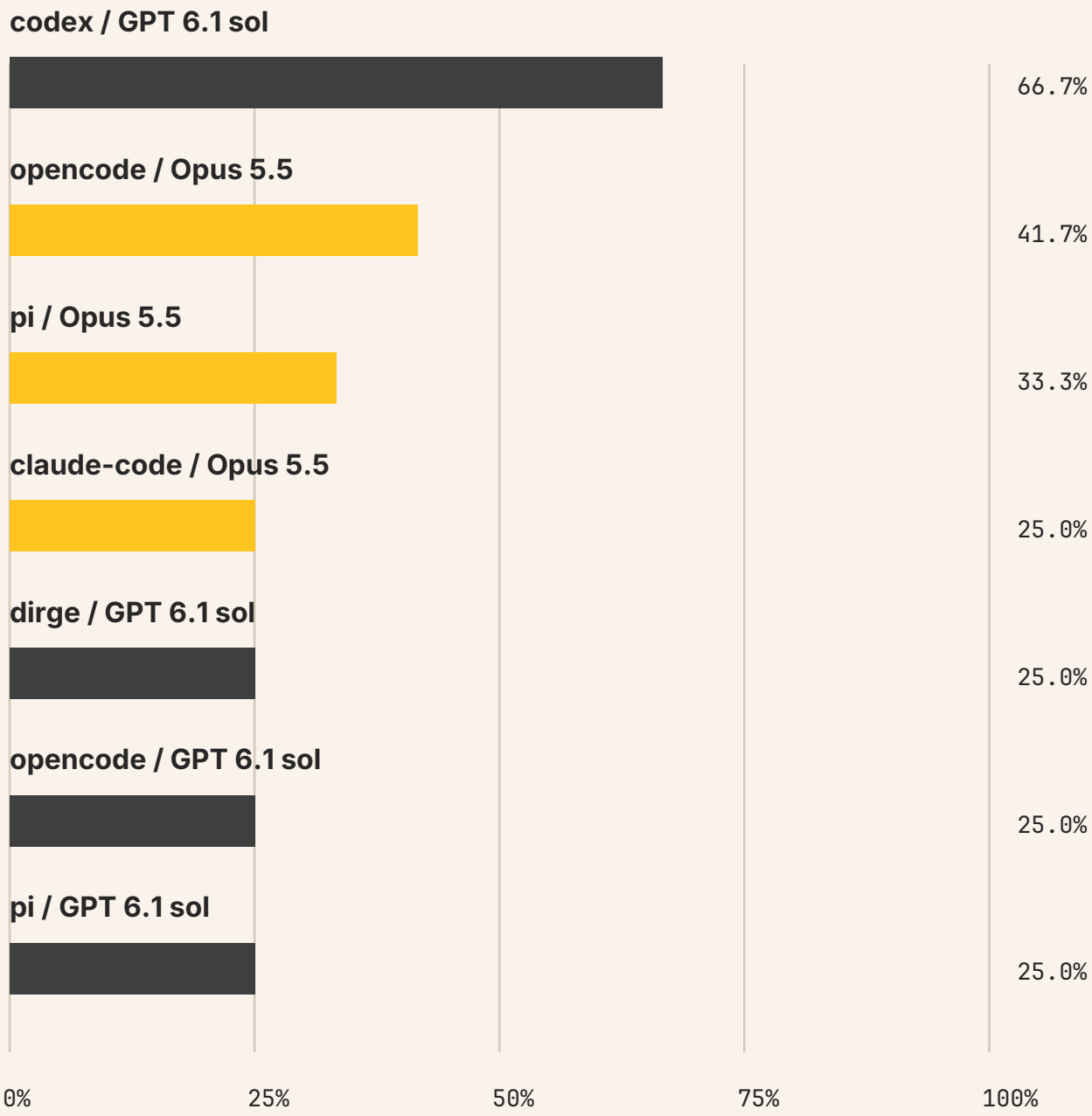
OpenCode on Opus uses a 128,000-token output cap. Other configurations retain their recorded defaults. Cost uses agent-reported values where available, otherwise pinned API prices. Time covers agent execution, excluding setup and verification.

## Evidence and exclusions

GPT and Claude Code results use sealed r3. The later pi and OpenCode Opus trials have hash-locked evidence from a dirty checkout, outside the clean-commit seal. Two OpenCode setup failures are retained and replaced: html-js-filter scored 0 and photonic-waveguide-routing scored 1. The earlier 32k Opus smoke is excluded.

# Task resolution

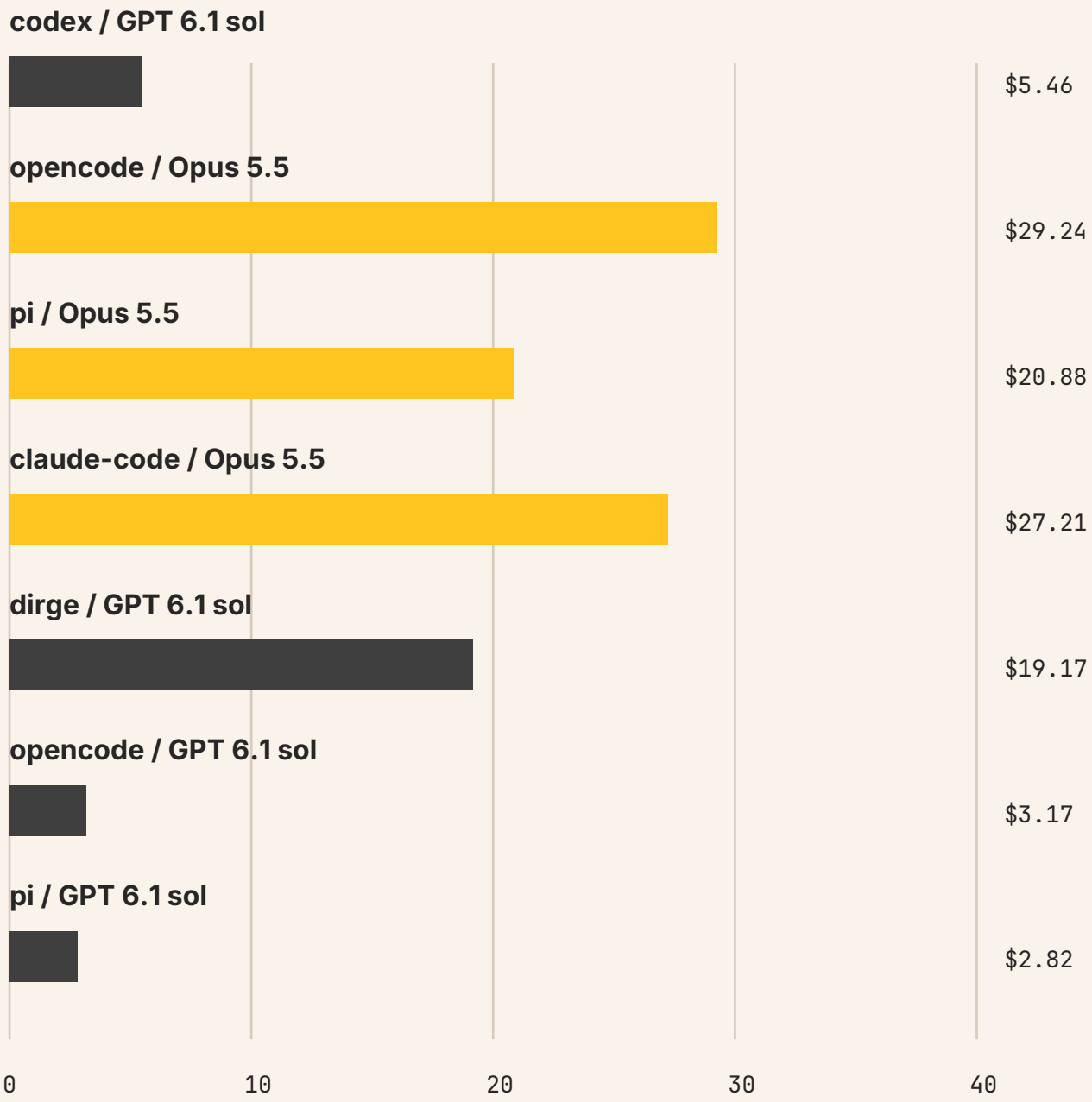
Resolved tasks / 12 (%). Amber: Opus 5.5. Graphite: GPT 6.1 sol.




Codex / GPT resolves 8 tasks; OpenCode / Opus resolves 5; pi / Opus resolves 4. The other pairs resolve 3 each.

# Cost ($)

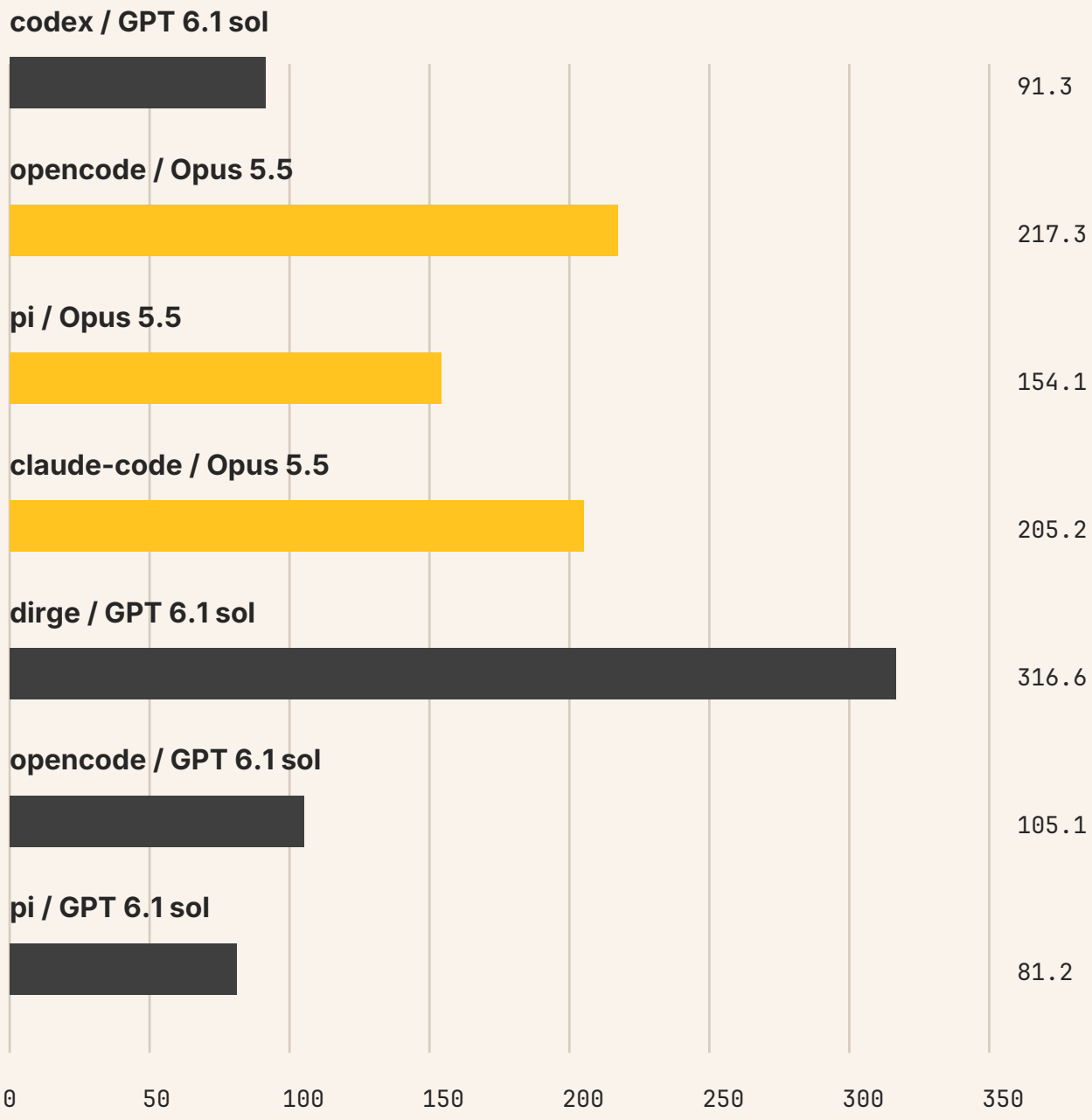
Total cost ($) across twelve scored attempts per pair.




Pi / GPT has the lowest measured cost at $2.82. Codex / GPT resolves the most tasks at $5.46.

# Execution time

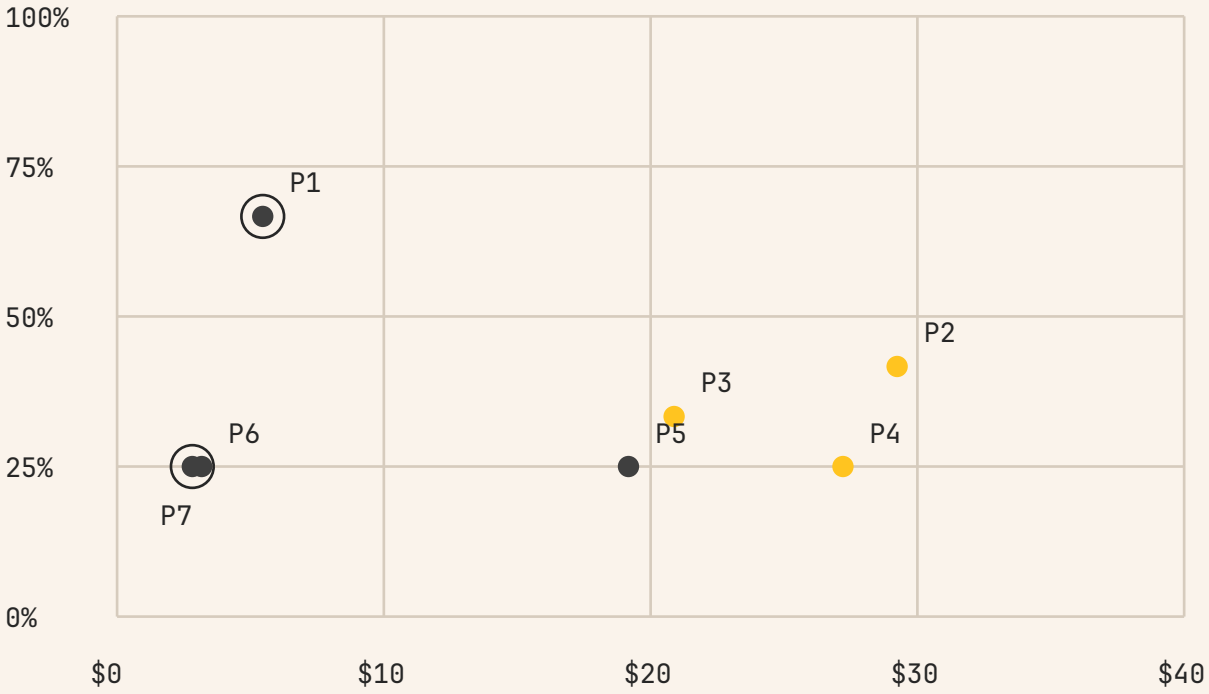
Total agent execution time (minutes) across twelve scored attempts per pair.




Totals exclude environment setup, verifier execution and excluded attempts.

# Cost and resolution

Resolution rate (%) against cost ($). Rings mark non-dominated pairs.




P1 codex / GPT 6.1 sol

P2 opencode / Opus 5.5

P3 pi / Opus 5.5

P4 claude-code / Opus 5.5

P5 dirge / GPT 6.1 sol

P6 opencode / GPT 6.1 sol

P7 pi / GPT 6.1 sol

The observed frontier contains pi / GPT and Codex / GPT. A non-dominated pair has no alternative that costs no more and resolves at least as many tasks, with one strict improvement.

# Metrics by configuration

Totals across 12 tasks. Input includes cached tokens; time is agent execution.

### P1 Harness: codex

Model: GPT 6.1 sol

Input tokens: 17,803,075

Output tokens: 238,659

Time: 91.3 min

Cost: $5.46

Resolved: 8/12

### P2 Harness: opencode

Model: Opus 5.5

Input tokens: 38,765,614

Output tokens: 291,232

Time: 217.3 min

Cost: $29.24

Resolved: 5/12

### P3 Harness: pi

Model: Opus 5.5

Input tokens: 21,789,346

Output tokens: 609,789

Time: 154.1 min

Cost: $20.88

Resolved: 4/12

### P4 Harness: claude-code

Model: Opus 5.5

Input tokens: 37,137,069

Output tokens: 617,645

Time: 205.2 min

Cost: $27.21

Resolved: 3/12

# Metrics by configuration

Totals across 12 tasks. Input includes cached tokens; time is agent execution.

### P5 Harness: dirge

Model: GPT 6.1 sol

Input tokens: 12,348,328 Output tokens: 159,253

Time: 316.6 min Cost: $19.17

Resolved: 3/12

### P6 Harness: opencode

Model: GPT 6.1 sol

Input tokens: 11,952,272 Output tokens: 90,545

Time: 105.1 min Cost: $3.17

Resolved: 3/12

### P7 Harness: pi

Model: GPT 6.1 sol

Input tokens: 5,908,447 Output tokens: 149,353

Time: 81.2 min Cost: $2.82

Resolved: 3/12

# Task-level outcomes

1 = resolved. 0 = unresolved. * = runtime exception or provider refusal.

| | P1 | P2 | P3 | P4 | P5 | P6 | P7 |
|---|---|---|---|---|---|---|---|
| bun-sourcemap-leak | 0 | 0 | 0 | 0 | 0* | 0 | 0 |
| html-js-filter | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| interleaved-vigenere | 0 | 1 | 1 | 0* | 1 | 0 | 0 |
| kv-live-surgery | 1 | 0* | 0 | 0 | 0 | 0 | 0 |
| music-harmony | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| mvcc-lsm-compaction | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| payments-pipeline-fix | 1 | 1 | 1 | 1 | 0* | 0 | 1 |
| photonic-waveguide-routing | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| react-lead-form | 0 | 1 | 1 | 1 | 0* | 0 | 0 |
| session-window-debug | 1 | 0 | 0 | 0 | 0* | 0 | 0 |
| vf2-speedup-networkx | 1 | 1 | 0 | 0 | 0 | 0* | 0 |
| wal-recovery-ordering | 1 | 0 | 0 | 0 | 0 | 1 | 1 |

Provider refusals occurred for Claude Code on interleaved-vigenere and OpenCode Opus on kv-live-surgery. OpenCode GPT hung on vf2-speedup-networkx; its cause remains unresolved.

# Interpretation

## Completion and cost

Codex / GPT leads completion with 8/12 resolved tasks at $5.46. Pi / GPT has the lowest total cost at $2.82 and resolves 3/12. These pairs define the observed cost–resolution frontier.

## Within-harness differences

OpenCode resolves 5/12 tasks with Opus and 3/12 with GPT. Pi resolves 4/12 with Opus and 3/12 with GPT. The higher Opus scores come with higher costs. These differences include model, plugin and budget effects.

## Task sensitivity

One task changes the resolution rate by 8.3 percentage points. No configuration resolves bun-sourcemap-leak or music-harmony. Repeated runs can distinguish stable failure patterns from run-to-run variation.

## Use of the results

For this workload, Codex / GPT offers the highest completion rate, while pi / GPT offers the lowest measured cost. Tool selection can use these two reference points alongside the task-level results.

# References

## Study records

Task selection: tasks.toml. Scored outcomes and aggregate metrics: results/opus-comparison/data.json. Evidence hashes: results/opus-comparison/manifest.json.

Presentation assets and PDF hashes: results/opus-comparison-branded/manifest.json. The report uses the same 72 scored attempts as the source comparison.

## Evaluation methodology

Anthropic — SWE-bench engineering report. Reference for evaluating a model together with its harness.

https://www.anthropic.com/engineering/swe-bench-sonnet

Terminal-Bench — leaderboard presentation of task resolution and uncertainty. The fetched page at the link below displayed Terminal-Bench 4.0.

https://www.tbench.ai/leaderboard/terminal-bench/2.0

## Brand assets

Blockether — typeface, symbol and wordmark geometry from https://blockether.com/. The source font and measured geometry are retained with the presentation assets.

# Further work

## 01 / Repeat the trials

Run multiple attempts for each pair and task. Report variation in completion, cost and time, then estimate uncertainty from the repeated measurements.

## 02 / Match the budgets

Align output caps, reasoning settings and time limits where the harnesses permit. Record remaining configuration differences alongside each result.

## 03 / Expand task coverage

Add tasks across difficulty levels and domains. Define the selection procedure before execution and retain task-level outcomes for each configuration.

## 04 / Resolve failure mechanisms

Reproduce the OpenCode GPT hang on vf2-speedup-networkx. Inspect verifier artifacts for the shared failures on bun-sourcemap-leak and music-harmony.

## 05 / Standardize the next run

Run all configurations from a clean, pinned checkout in one campaign. Use a common cost ledger and apply the standard evidence seal to every trial.